

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:BE01000051

Date:06-01-2025

Subject Name:Basic Electrical Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Apply and illustrate KCL on star connected circuit.	03
	(b) Compute R_{th} and V_{th} for thevenin theorem. Assume necessary circuit for the same.	04
	(c) Explain star to delta conversion with neat and clean diagram.	07
Q.2	(a) Explain the concept of active, reactive and apparent power.	03
	(b) Analyze voltage and current relations in three phase star connection.	04
	(c) Compare series and parallel resonance.	07
	OR	
	(c) Illustrate the Phasor diagram of RC series circuit and explain its development.	07
Q.3	(a) Describe concept of MMF.	03
	(b) Define flux, flux density, reluctance and permeability.	04
	(c) Compare Split phase and Capacitor start motors.	07
	OR	
Q.3	(a) Describe Construction and working of brushless DC motor.	03
	(b) Examine construction of single phase induction motor.	04
	(c) Explain B-H Curve and Hysteresis Loop in detail.	07
Q.4	(a) List components of low tension switchgear.	03
	(b) Define types of lamps for industrial and residential premises.	04
	(c) Explain Lumen requirements for different categories.	07
	OR	
Q.4	(a) Describe working of MCB.	03
	(b) State different parts with neat and clean diagram of MCCB.	04
	(c) Explain working principal of energy meter.	07
Q.5	(a) List various methods of earthing.	03
	(b) Describe the necessity of using electrical protective devices.	04
	(c) Explain Pipe earthing.	07
	OR	
Q.5	(a) State important characteristics of batteries.	03
	(b) State safety precautions of electrical appliances.	04
	(c) Explain plate earthing.	07

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000051

Date:13-06-2025

Subject Name:Basic Electrical Engineering

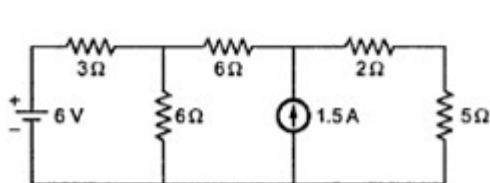
Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) State Ohm's law and Kirchhoff's laws in context with DC circuits.	03
(b) Compare resistive series and parallel circuit.	04
(c) Find the current through 5 ohm resistance using the superposition theorem.	07



Q.2 (a) Define Q factor of series RLC circuit. What is the importance of it?	03
(b) Draw impedance triangle, voltage triangle and power triangle for single phase R-L circuit.	04
(c) Prove that current in purely capacitive circuit current leads its voltage by 90 and average consumption in pure capacitive circuit is 0.	07

OR

(c) A single phase R-L-C circuit having resistance of 8 Ω, inductance of 80 mH and capacitance of 100 μF is connected across single phase ac 150 V, 50 Hz supply. Calculate the current, power factor and voltage drop across inductance and capacitance.	07
Q.3 (a) Explain magnetic hysteresis.	03
(b) Explain working principle of single phase Transformer.	04
(c) Explain with diagram construction of core type and shell type transformer.	07

OR

(a) Derive the E.M.F. equation of a single phase transformer.	03
(b) Explain the terms:- 1. Flux density 2. Reluctance 3. Permeability 4. Electric field strength.	04
(c) State comparison between practical transformer and ideal transformer.	07

- Q.4 (a)** Give comparison between Fuse and MCB. **03**
- (b)** Mention the types of single phase induction motor. Explain any one of them. **04**
- (c)** Give merit, demerit and application of induction motors. **07**

OR

- (a)** Explain the working principle of single phase induction motor. **03**
- (b)** Explain why single-phase induction motors are commonly used in household appliances like fans and refrigerators. **04**
- (c)** Explain generation of rotating magnetic field in 1 – phase induction motor with diagram and equations. **07**
- Q.5 (a)** Write short note on ELCB. **03**
- (b)** Write safety precaution for electrical appliances. **04**
- (c)** What is battery? Explain the construction and working of any battery. **07**

OR

- (a)** Give a list of safety device used for electrical appliances. **03**
- (b)** What is need of earthing ? Explain the different method of earthing. **04**
- (c)** Calculate the electricity bill amount for a month of April, if 4 bulbs of 40 W for 5 h, 4 tube lights of 60 W for 5 h, a TV of 100 W for 6 h, a washing machine of 400 W for 3 h, a water pump of 0.5 HP for 15 minutes are used per day. The cost per unit is Rs 3.50. Consider 1 HP = 746 watts. **07**

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000051/BE01R00051

Date: 24-12-2025

Subject Name: Basic Electrical Engineering

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

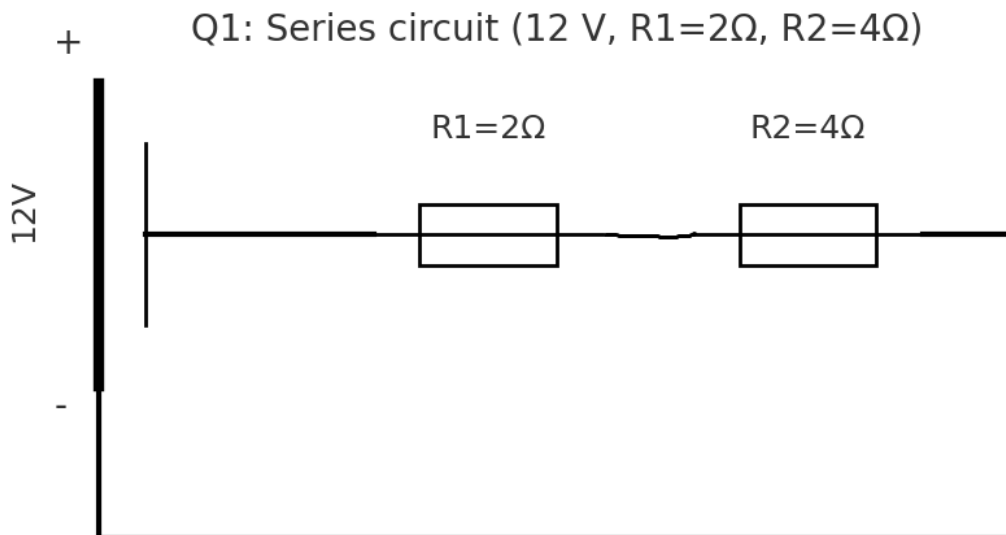
Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Marks

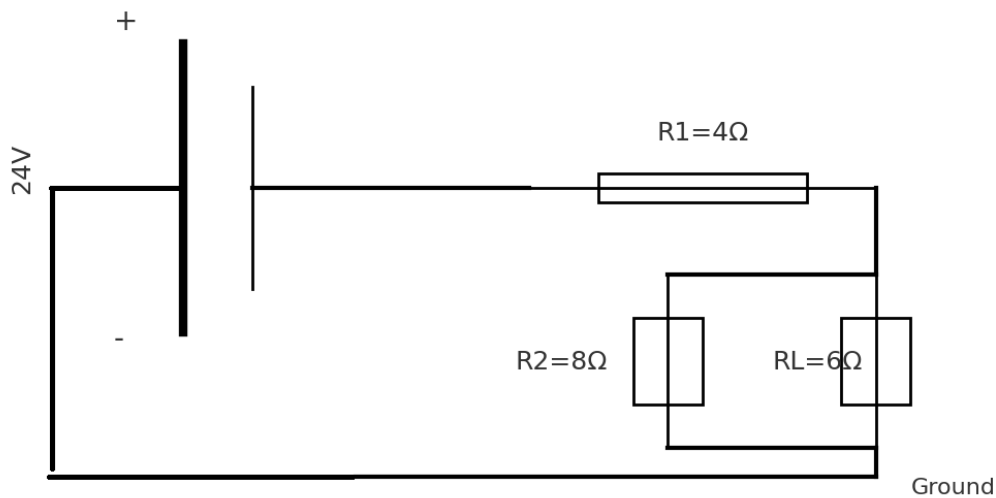
- Q.1 (a)** Using Kirchhoff's Voltage Law (KVL), calculate the current flowing through a $2\ \Omega$ resistor in a simple series circuit containing a 12 V battery, a $2\ \Omega$ resistor, and a $4\ \Omega$ resistor.

03



- (b) For the given circuit, apply Thevenin's Theorem to determine the current flowing through a $6\ \Omega$ load resistor connected across two terminals. (Assume: a 24 V source with $4\ \Omega$ and $8\ \Omega$ resistors in series, load of $6\ \Omega$ connected across the $8\ \Omega$ resistor terminals.)

04



- (c) A DC circuit consists of three resistors: $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, and $R_3 = 30\ \Omega$, connected in a star configuration. Convert this star network into its equivalent delta network, and calculate the resistance values of each branch of the delta.

07

- Q.2** (a) Compare the behavior of a pure resistor, a pure inductor, and a pure capacitor when each is connected to a single-phase AC supply.

03

- (b) Analyze and explain why the power factor in an inductive load is always lagging, while in a capacitive load it is leading.

04

- (c) A three-phase system can be connected in either star (Y) or delta (Δ) configuration. Analyze and discuss the differences between these two connections in terms of:

07

- Line and phase voltages
- Line and phase currents
- Typical applications in electrical systems

OR

- (c) Analyze the conditions for series resonance in a single-phase RLC circuit. Discuss how:

07

- Voltage and current behave at resonance,
- Power factor changes at resonance, and
- Two practical applications of resonance in electrical systems.

- Q.3** (a) Explain the difference between a magnetic circuit and an electric circuit with one simple example each.

03

- (b) Describe the working principle of a single-phase transformer. Why is it called a “static” electrical machine?

04

- (c) Explain the construction and working of a single-phase induction motor. 07
Also, list at least three common household applications where this motor is used.

OR

- (a) What is hysteresis loss in magnetic materials? Explain why it occurs in transformers. 03
(b) Describe the term rotating magnetic field. How is it produced in an induction motor? 04
(c) Explain the working principle of a brushless DC motor (BLDC). Also, mention two 07
advantages of BLDC motors over traditional DC motors.

- Q.4** (a) Explain the difference between MCB (Miniature Circuit Breaker) and ELCB (Earth 03
Leakage Circuit Breaker).
(b) Describe the importance of earthing in domestic wiring systems. Mention two 04
common types of earthing.
(c) Explain the general procedure of preparing an electricity bill for a household. Also, 07
discuss how energy consumption of appliances is calculated in kilowatt-hours
(kWh).

OR

- (a) What are the different types of lamps used in domestic and commercial wiring? 03
Briefly explain any two.
(b) Explain the function of a wattmeter. How is it different from an ammeter and a 04
voltmeter in electrical measurements?
(c) Explain the safety precautions that should be taken while handling household 07
electrical appliances. Support your answer with at least three examples.

- Q.5** (a) State the working principle of a voltmeter. 03
(b) State the characteristics of a battery. 04
(c) Write short note on Switch Fuse Unit (SFU) 07

OR

- (a) State any three applications of brushless DC (BLDC) motors in daily life. 03
(b) State different types of batteries and mention one example of each. 04
(c) Write short note on Earth Leakage Circuit Breaker (ELCB) 07
